TANG XINYI (唐鑫夷)

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EDUCATION

Beijing Normal University, Beijing, China 2023 – 2026
Master of Arts, School of Arts and Media
Supervisor: Prof. Lun Zhang
Thesis: Imagining the Media Roles of Conversational AI: A Computational Grounded Theory Study with Multi-Agent Frameworks

Communication University of China, Beijing, China 2019 – 2023
Bachelor of Engineering, School of Information and Communication Engineering
Academic supervisor in the Excellence Program: Prof. Fulian Yin
Thesis: Dynamics of information propagation and intervention in the multi-platform coupled networks

HONORS & AWARDS

First-Class Academic Scholarship, Beijing Normal University 2025
Outstanding Graduate in Beijing, China 2023
Outstanding Graduate at Communication University of China 2023
Second Prize, National Mathematical Modeling Contest (Beijing Division) 2021
Multiple scholarships, Communication University of China 2019 – 2023
Including First-Class Academic Scholarship, Outstanding Student Leader Scholarship, etc.

RESEARCH INTERESTS

Human-AI Social Interaction

- I study how users develop attitudes toward AI agents and how these attitudes give rise to new forms of human-AI relationships.
- Rooted in my earlier work on public opinion dynamics on social media, I examine the motivations, cultural conditions, and emergent relational forms in these practices, with implications for the design of socially-aware AI systems.

AI-Augmented Social Science Methodology

- I develop mixed-method approaches integrating computational techniques with qualitative inquiry, and critically examine AI's potential role as an epistemic partner in the research process.
- This interest began with experiments in LLM-assisted content analysis and has since extended to multi-agent frameworks for computational grounded theory.

RELEVANT RESEARCH SKILLS

Qualitative Methods: Grounded theory method, semi-structured interviewing
Computational Methods: NLP, LLM-based multi-agent system design, data mining, statistical analysis
Programming & Tools: Python, R, HTML/CSS/JavaScript, Vibe Coding

TEACHING EXPERIENCE

Teaching Assistant: *Vibe Coding with AI*, Beijing Normal University 2024 – 2025

RESEARCH EXPERIENCE

Human-AI Social Interaction on Social Media 2023 – 2025

Principal Investigator

- Investigated how users perceive and construct social relationships with conversational AI agents
- Conducted semi-structured interviews to explore early adopters' practices in human-AI interactions
- Extended the inquiry through large-scale analysis of user-generated content on Xiaohongshu (RedNote), integrating computational techniques with grounded theory methodology
- Designed an LLM-based multi-agent framework for computational grounded theory

Data-Driven Simulation of Public Opinion Dynamics on Social Media 2021 – 2023

Project Leader, National Innovation and Entrepreneurship Training Program

- Built simulation models of public opinion propagation across multiple social media platforms
- Modeled coupled dynamics of information diffusion and emotional contagion during public emergencies
- Investigated intervention strategies for opinion governance
- Developed methods for evaluating propagation patterns, resulting in three national invention patents

INDUSTRY & PRODUCT EXPERIENCE

China Telecom System Integration Co., Ltd. 2023

Product Manager Intern

- Managed big data products in the public security sector, supporting government agencies and courts in social risk governance
- Authored product requirement documents and designed product prototypes for public opinion risk monitoring and early warning systems

National Key R&D Program of China (No. 2021YFF0901705) 2021 – 2023

Core R&D Member

Sub-project: *Application and Demonstration of Cultural Resource Reuse and Big Data Services*

- Conducted industry analysis of digital cultural products, surveying market landscape and practices
- Authored the design proposal for film and television digital cultural products
- Collected film and television data through web scraping and performed data mining and analysis

PUBLICATIONS

Conference Papers

[1] **Tang, X. Y.**, & Zhang, L. Imagining the Media Roles of Conversational AI: A Computational Grounded Theory Study with Multi-Agent Frameworks. Target: *CHI 2027*. (In preparation)

[2] **Tang, X. Y.**, Lu, J. L., Zhang, L., Wu, Y., & Xu, X. K. (2025). Aligning AI With What: Role Coordination through Perception, Construction, and Adaptation during Human-AI Interactions. Paper presented at the Human-Machine Communication Division, *2025 ICA Annual Conference.* Denver, USA.

[3] **Tang, X. Y.**, & Zhang, L. (2024). Competition, symbiosis and collaborative attenuation: Paradigm transformation of journalism and communication in the 5G era. *2024 Annual Conference of Intelligent and Computational Communication*, Zhejiang, China.

[4] Wang, J. X., Liang, T. Y., Kuang, Q. H., **Tang, X. Y.**, Ma, R., & Yin, F. L. (2022). Combined influence of commenting and forwarding on information propagation on the Chinese Sina Microblog. *2022 7th International Conference on Cloud Computing and Big Data Analytics*, Chengdu, China, pp. 342–347.

Journal Articles

[1] **Tang, X. Y.**, & Zhang, L. (2026). Paradigmatic Integration, Consolidation, and Differentiation: Information Infrastructure and the Paradigm Shifts of Journalism and Communication Studies. *Journalism & Communication Studies*. **(Under review, 3rd round, CSSCI TOP)**

[2] Yin, F. L., Wang, J. X., **Tang, X. Y.**, Jiang, X. Y., Wu, Y. W., & Wu, J. H. (2025). Dynamics of information propagation and intervention in the multiplatform coupled networks. *Expert Systems with*

Applications, 277, 127262. **(SCI TOP)**

[3] Yin, F. L., **Tang, X. Y.**, Liang, T. Y., Kuang, Q. H., Wang, J. X., Ma, R., ... & Wu, J. H. (2024). Coupled dynamics of information propagation and emotion influence: Emerging emotion clusters for public health emergency messages on the Chinese Sina Microblog. *Physica A: Statistical Mechanics and its Applications*, 639, 129630. **(SCI Q2)**

[4] Yin, F. L., She, Y. W., Pan, Y. Y., **Tang, X. Y.**, Hou, H. T., & Wu, J. H. (2023). Hot-topics cross-propagation and opinion-transfer dynamics in the Chinese Sina-microblog social media: A modeling study. *Journal of Theoretical Biology*, 566, 111480. **(SCI Q3)**

[5] Yin, F. L., **Tang, X. Y.**, Liang, T. Y., Huang, Y. J., & Wu, J. H. (2022). External intervention model with direct and indirect propagation behaviors on social media platforms. *Mathematical Biosciences and Engineering*, 19(11), 11380–11398. **(SCI Q2)**

[6] Yin, F. L., Pan, Y. Y., **Tang, X. Y.**, Wu, C., Jin, Z., & Wu, J. H. (2022). An information propagation network dynamic considering multi-platform influences. *Applied Mathematics Letters*, 133, 108231. **(SCI TOP)**

[7] Yin, F. L., Wu, Z. L., Shao, X. Y., **Tang, X. Y.**, Liang, T. Y., & Wu, J. H. (2022). Topic—a cluster of relevant messages—propagation dynamics: A modeling study of the impact of user repeated forwarding behaviors. *Applied Mathematics Letters*, 127, 107819. **(SCI TOP)**

Patents

[1] Information propagation analysis method and system based on social media pseudo-environment modeling. National Invention Patent No. ZL202210645833.8. Granted 2022.

[2] Hybrid information propagation dynamics model and its information propagation analysis method. National Invention Patent No. ZL202210218560.9. Granted 2022.

[3] Public topic propagation evaluation method and system based on modeling. National Invention Patent No. ZL202110812031.7. Granted 2021.

ACADEMIC SERVICE

Peer Reviewer: *Information Processing & Management* (SCI TOP), *ICA 2025*